

Reducing Hallucinations via Citation-Grounded QA with Locally-Hosted Small Language Models

CS505 Natural Language Processing | Final Project Proposal | Boston University

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Project type: Empirical NLP project focused on retrieval-augmented generation (RAG) for hallucination reduction.

Research question: *Within a fully local, privacy-preserving RAG system, how much of a sub-7B SLM's hallucination rate is attributable to retrieval quality versus generation capacity?*

1 Problem Statement & Motivation

Hallucination generating claims unsupported by source documents is the primary failure mode of LLMs in high-stakes settings. Cloud tools like Google's NotebookLM require transmitting corpora to external servers, conflicting with HIPAA/GDPR requirements. **Cloud upload equals data exfiltration risk.** Sub-7B SLMs (e.g., Phi-3-mini-3.8B [Microsoft, 2024]) can run on a single consumer GPU, making local deployment feasible.

Task definition. Given a corpus $\mathcal{C} = \{d_1, \dots, d_n\}$ and query q , the system produces answer a plus, for every sentence $s_i \in a$, a citation (p_i, s_i) where passage $p_i \in \mathcal{C}$ textually entails s_i . Any sentence without a valid citation is counted as a hallucination.

2 Proposed System

All components run locally; no external API calls are required.

Parsing. QASPER provides structured JSON with section boundaries, which we load directly.

Chunking (ablated). Three strategies: fixed-size (256 tok, 32 overlap), sliding window (512 tok, 128 overlap), and section-aware (boundary-aligned, capped at 512 tokens).

Retrieval (ablated). Dense only (FAISS + BGE-small-en-v1.5); BM25 (rank-bm25); hybrid reciprocal-rank fusion (RRF); hybrid + cross-encoder reranking (ms-marco-MiniLM-L-6-v2), top-5 passages.

Generation. Two prompt variants: standard generation vs. *citation forcing*, where the model must bracket every claim with an inline citation and assert only what the retrieved passages support. Served via Ollama; ≤ 16 GB VRAM.

3 Experimental Design

3.1 Conditions

Each condition changes exactly one component (Table 1); F and G are stretch goals.

Cond.	Chunk	Retrieval	Rerank	Cite force	Generator
A (base)	Fixed	Dense	×	×	Phi-3-mini
B	Fixed	Hybrid	×	×	Phi-3-mini
C	Section	Hybrid	×	×	Phi-3-mini
D	Section	Hybrid	✓	×	Phi-3-mini
E (best)	Section	Hybrid	✓	✓	Phi-3-mini
F*	Section	Hybrid	✓	✓	Llama-3-8B
G*	Section	Hybrid	✓	✓	Mistral-7B

Table 1: Ablation conditions. *Stretch goals (time permitting).

3.2 Dataset & Baselines

Dataset: QASPER [Dasigi et al., 2021] 1,585 NLP papers, 5,049 Q&A pairs with gold evidence annotations supporting both retrieval and answer evaluation (official token F1).

Baselines: (1) Parametric-only Phi-3-mini; (2) BM25-only; (3) Random retrieval (5 passages).

3.3 Metrics & Hypothesis

Primary: Citation precision via NLI entailment (TRUE cross encoder) and hallucination rate. **Secondary:** Token F1 (QASPER official), BERTScore. **Faithfulness:** RAGAS [Es et al., 2023] (Llama-3-8B local judge).

Retrieval: Recall@5, MRR@10, evidence hit rate.

Hypothesis: Retrieval improvements (A→D) will account for the majority of hallucination reduction, with citation forcing (D→E) providing a smaller incremental gain; results will be stratified by whether gold evidence was retrieved to attribute failures to retrieval vs. generation.

4 Feasibility & Timeline

Wk	Milestone
1–2	Parsing, chunking strategies, BM25 + dense baseline (cond. A)
3	Hybrid RRF + reranking; conditions B to D; QASPER eval pipeline
4	Citation forcing (E); RAGAS + NLI evaluation; error analysis
5	Midway report; begin stretch goals (F, G) if on schedule
6	Final ablation, writeup (ACL format)

Table 2: Six week schedule. Core scope: weeks 1–4 (conditions A–E).

Stack: sentence-transformers, FAISS, rank-bm25, Ollama, RAGAS, HuggingFace Transformers; no paid API for core scope.

References

- P. Dasigi, K. Lo, I. Beltagy, A. Cohan, N. A. Smith, and M. Gardner. A dataset of information-seeking questions and answers anchored in research papers. In *Proceedings of NAACL*, 2021.
- S. Es, J. James, L. Espinosa-Anke, and S. Schockaert. RAGAS: Automated evaluation of retrieval augmented generation. *arXiv preprint arXiv:2309.15217*, 2023.
- Microsoft. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024.